

Introduction to Microeconomics

11–Profit Maximization, Supply and Market Equilibrium in Perfectly Competitive Markets

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ECON101
Spring 2026

Perfectly Competitive Market

A perfectly competitive market is characterized by

- Many small firms compared to the total size of the market.
- Homogeneous product.
- Full information.
- Negligible Transaction Costs.
- Free Entry and Exit.

We will develop the theory of the firm operating in a perfectly competitive market. However, many of the insights that we develop here extends to cases in which some of the characteristics above are not true. Turns out that none of the above are essential for a market to be “competitive”. What is important is that the market has a **contestable structure: low or nonexistent sunk costs, and the possibility to use strategies to undercut rivals, i.e., price adjustments by the firms are felt strongly by consumers.**

Revenues

Given a market demand curve $Q = D(p)$ with inverse demand curve $p = D^{-1}(Q)$, if the firm faces this demand curve (in full) revenues for the firm are defined as:

$$R(Q) = \text{Price} \times \text{Quantity} = D^{-1}(Q) \times Q.$$

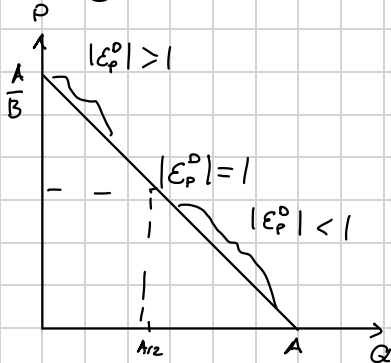
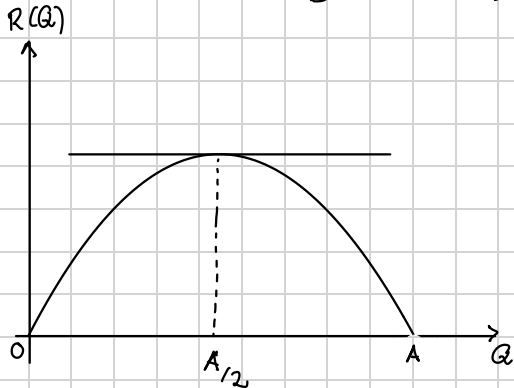
For example, if $Q = A - Bp$, the inverse demand is $p = A/B - Q/B$, thus

$$\begin{aligned} R(Q) &= \left(\frac{A}{B} - \frac{Q}{B} \right) Q \\ &= \frac{A}{B} Q - \frac{Q^2}{B} \end{aligned}$$

Revenues

$$R(Q) = \left(\frac{A}{B} - \frac{Q}{B}\right)Q$$

$$O = \left(\frac{A}{B} - \frac{Q}{B}\right)Q$$



$$D(P) = A - BP$$

$$\rightarrow R'(Q) = \frac{dR(Q)}{dQ} = \frac{A}{B} - \frac{2Q}{B}$$

$$R'(Q_{max}) = 0 \Leftrightarrow Q_{max} = \frac{A}{2}$$

Profit Maximization

The firm's profit function, for any q , is

economic profit $\rightarrow \Pi(q) = R(q) - C(q)$ *economic costs = explicit + implicit*

where $R(q)$ are revenues and $C(q)$ are costs as a function of quantity.

The firm's problem is then to **maximize profits** by choosing optimally the quantity offered.

$$\max_q R(q) - C(q) = \max_q \pi(q)$$

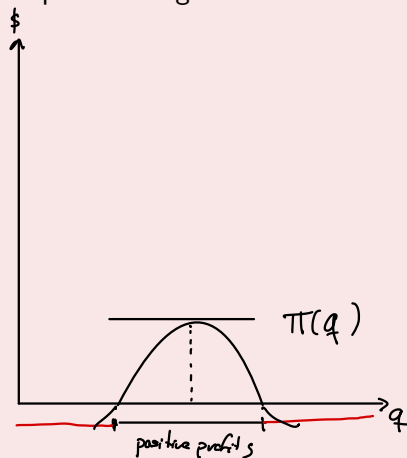
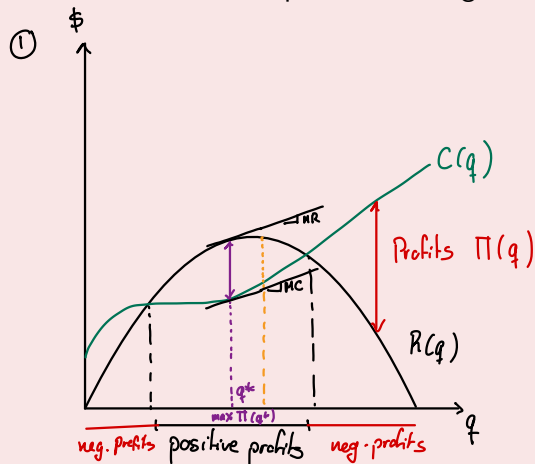
economic profits

There are two steps in this problem:

- Output decision. What output level q^* maximizes profit or minimizes loss?
- Shutdown decision. Is it more profitable to produce q^* or to shutdown and produce no output?

Output rules

- 1 The firm sets its output where its profit is maximized.
- 2 The firm sets its output where marginal profit is equal to zero.
- 3 The firm sets its output where marginal revenue is equal to marginal cost.



$$(2) \quad \Pi(q) = R(q) - C(q)$$

$$\text{Marginal profit} = \frac{d\Pi(q)}{dq} = \frac{dR(q)}{dq} - \frac{dC(q)}{dq} \stackrel{\text{F.O.C.}}{=} 0 \quad \therefore \text{Marginal Profit}(q^*) = 0$$

$$\Rightarrow \frac{dR(q^*)}{dq} = \frac{dC(q^*)}{dq}$$

$$(3) \quad \text{MR}(q^*) = \text{MC}(q^*) \quad \therefore \text{Marginal Revenue at } q^* = \text{Marginal Cost } q^*$$

↳ $\text{MR}(q) > \text{MC}(q)$: Firm can increase profits by increasing q

$\text{MR}(q) < \text{MC}(q)$: Firm can increase profits by decreasing q

Shutdown rules

- 1 The firm shuts down only if it can reduce its loss by doing so.
- 2 The firm shuts down only if its revenue is less than its avoidable cost.

In the Short-Run : $C(q) = \underbrace{VC(q)}_{\substack{\text{Variable Costs} \\ (\text{depend on } q)}} + \underbrace{F}_{\text{Fixed Costs.}}$

← F : is unavoidable.
(sunk)

$$\Pi(q) = R(q) - VC(q) - F : \text{ Notice that } \frac{dC(q)}{dq} = \frac{dVC(q)}{dq}$$

$$q^* : MR(q^*) = MVC(q^*)$$

If $R(q^*) < VC(q^*) \Rightarrow$ Firm should shutdown. $\rightarrow \Pi(q^*) = \underbrace{R(q^*) - VC(q^*)}_{< 0} - F$

$\hookrightarrow \Pi(0) = -F : \bar{q} = 0$, not q^*

If $R(q^*) \geq VC(q^*)$ then the Firm should produce!

$\hookrightarrow \Pi(q^*) = \underbrace{R(q^*) - VC(q^*)}_{\geq 0} - F > -F = \Pi(0)$

Perfect Competition

Price-taking behavior

In a **competitive market**, firms are price takers.

- In both input and output markets.
- A price-taking firm faces a demand curve that is **horizontal at the market price**.
 - The individual firm faces a perfectly elastic demand curve.

The Residual Demand Curve

- Suppose Market demand is $D(p)$.
- Suppose that the (aggregate) supply curve
- Then the residual demand curve for an individual firm is

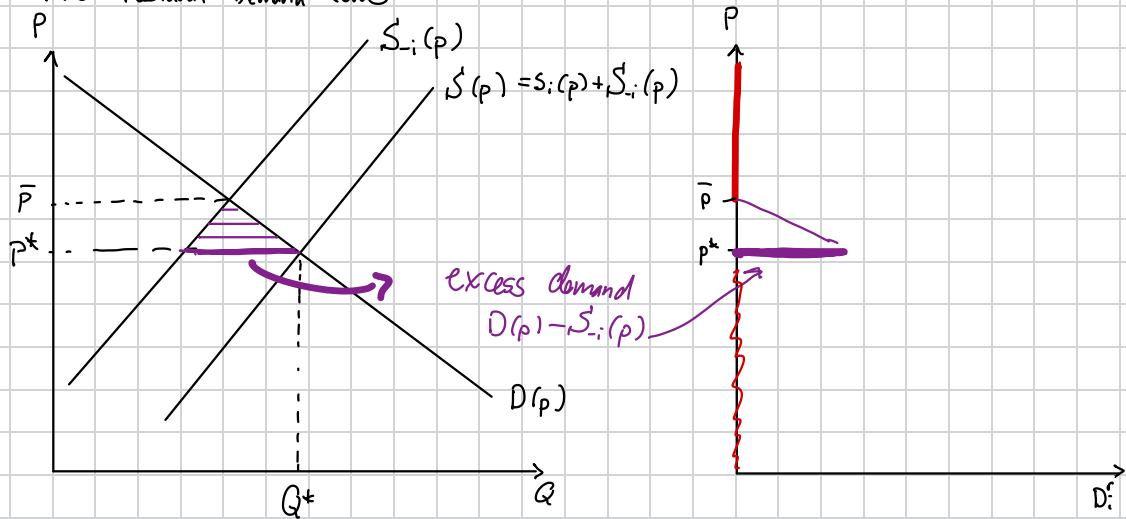
$$D_i^r(p) = D(p) - S_{-i}(p)$$

- What are the properties of $D_i^r(p)$?

Handwritten notes and equations:

$$S(p) = \overset{\text{Firm } i}{s_i(p)} + \overset{\text{Agg Firms other than } i}{S_{-i}(p)}$$
$$\sum_{j \neq i} s_j(p) = s_1(p) + s_2(p) + \dots + s_{i-1}(p) + \dots + s_{i+1}(p) + \dots + s_n(p)$$

The Residual Demand Curve



$D(p)$: market demand

$$S(p) = S_i(p) + S_{-i}(p)$$

$$D'_i(p) = D(p) - S_{-i}(p)$$

Remark Elasticity $\frac{\Delta\% y}{\Delta\% x} = \frac{\frac{\Delta y}{y}}{\frac{\Delta x}{x}} = \frac{\Delta y}{\Delta x} \cdot \frac{x}{y}$

$$y = f(x)$$

$$\Delta x \approx 0 \quad \epsilon_{y,x} = \frac{dy}{dx} \cdot \frac{x}{y}$$

$$\frac{d D'_i(p)}{d p} = D'(p) - S'_{-i}(p) \quad \epsilon'_{i,p} = (D'(p) - S'_{-i}(p)) \frac{p}{D'_i(p)}$$

$$\epsilon'_{i,p} = \frac{d D'_i(p)}{d p} \cdot \frac{p}{D'_i(p)}$$

$$\epsilon_D = D'(p) \frac{p}{D(p)}$$

$$\epsilon_{-i,S} = S'_{-i}(p) \frac{p}{S_{-i}(p)}$$

$$D'(p) = \frac{\epsilon_D D(p)}{p}, \quad S'_{-i}(p) = \frac{\epsilon_{-i,S} S_{-i}(p)}{p}$$

$$\epsilon'_{i,p} = \left(D'(p) - S'_{-i}(p) \right) \frac{p}{D'_i(p)}$$

$$\epsilon_D = D'(p) \frac{p}{D(p)} \quad \epsilon_{-i,S} = S'_{-i}(p) \frac{p}{S_{-i}(p)}$$

$$D'(p) = \frac{\epsilon_D D(p)}{p}, \quad S'_{-i}(p) = \frac{\epsilon_{-i,S} S_{-i}(p)}{p}$$

Plugging into $\epsilon'_{i,p}$:

$$\begin{aligned} \epsilon'_{i,p} &= \left(\epsilon_D \frac{D(p)}{p} - \epsilon_{-i,S} \frac{S_{-i}(p)}{p} \right) \frac{p}{D'_i(p)} \\ &= \epsilon_D \frac{D(p)}{D'_i(p)} - \epsilon_{-i,S} \frac{S_{-i}(p)}{D'_i(p)} \end{aligned}$$

→ Let p^* be the mkt. eq.
and at p^* let q_i be firm's i
production:

$$D(p^*) = q_i + S_{-i}(p^*)$$

$$D'_i(p^*) = q_i$$

$$\epsilon'_i(p^*) = \epsilon_D \frac{D(p^*)}{q_i} - \epsilon_{-i,S} \frac{S_{-i}(p^*)}{q_i}$$

$$\epsilon'_i(p^*) = \epsilon_D \frac{D(p^*)}{q_i} - \epsilon_{-i}^s \frac{S_{-i}(p^*)}{q_i}$$

Market Share of firm i:

$$s_i = \frac{q_i}{D(p^*)}$$

$$\frac{D(p^*)}{q_i} = \frac{1}{s_i} \quad S_{-i}(p^*) = (1-s_i) D(p^*)$$

$$\frac{S_{-i}(p^*)}{q_i} = \frac{1-s_i}{s_i}$$

So as $s_i \rightarrow 0$

$$|\epsilon'_i| \rightarrow \infty$$

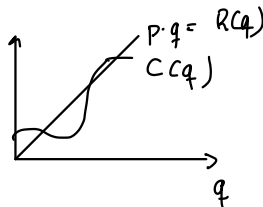
$$\epsilon'_i(p^*) = \frac{\epsilon_D}{s_i} - \epsilon_{-i}^s \frac{1-s_i}{s_i}$$

Competition in the Short Run & and the Short Run Output Decision

A firm under perfect competition, takes price as given, and faces a perfectly elastic demand at P .

Revenues $R(q) = P \cdot q$

↳ perfectly elastic
residual demand curve
firm faces at P



Short-run
 $C(0) = F$

$$C(q) = VC(q) + F$$

$$\pi(q) = P \cdot q - C(q)$$

Optimal output: $q^* : MR(q^*) = P = MC(q^*)$

$$MR = MC$$

$$P = MC(q^*)$$

$q^* > 0$ in order for the firm
to decide to produce

$$R(q^*) \geq VC(q^*) \Leftrightarrow P \cdot q^* \geq VC(q^*)$$

$$\Leftrightarrow P \geq \frac{VC(q^*)}{q^*} = AVC(q^*)$$

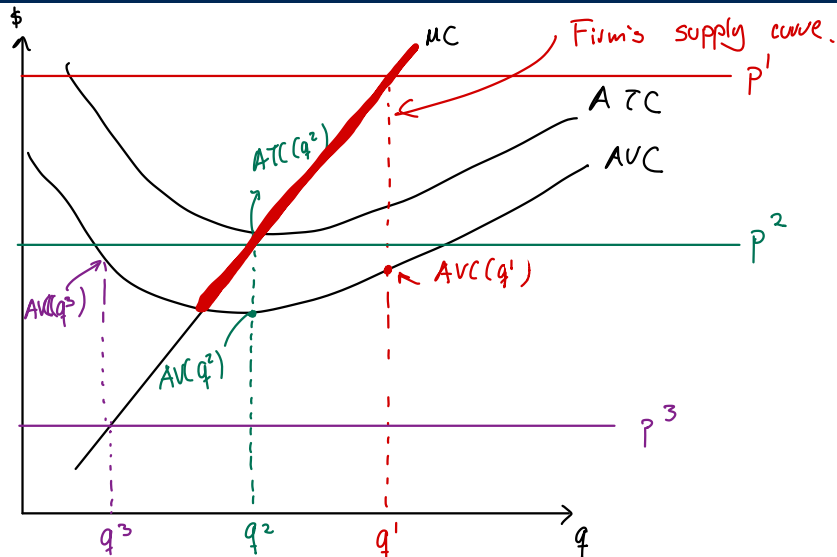
Firm shutdowns if $P < AVC(q^*)$

Competition in the Short Run & the Short Run Output Decision

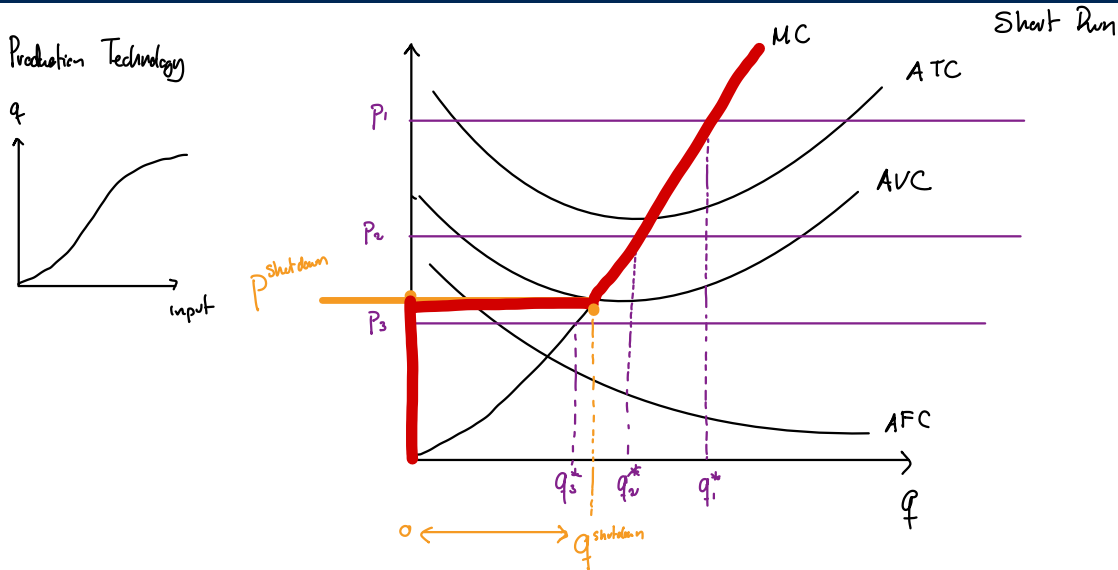
Firm produces at $q^* > 0$
if $p = MC(q^*)$
AND
 $p \geq AVC(q^*)$
otherwise it produces $\bar{q} = 0$

These expressions determine
the supply curve of the firm
under perfect competition:
 $p = MC(q)$ as long as
 $p \geq AVC(q)$
→ Inverse supply curve.

Competition in the Short Run & the Short Run Output Decision



Competition in the Short Run & the Short Run Output Decision



Competition in the Short Run & the Short Run Output Decision

$$AR = \frac{P \cdot q}{q} = P$$



Revenues



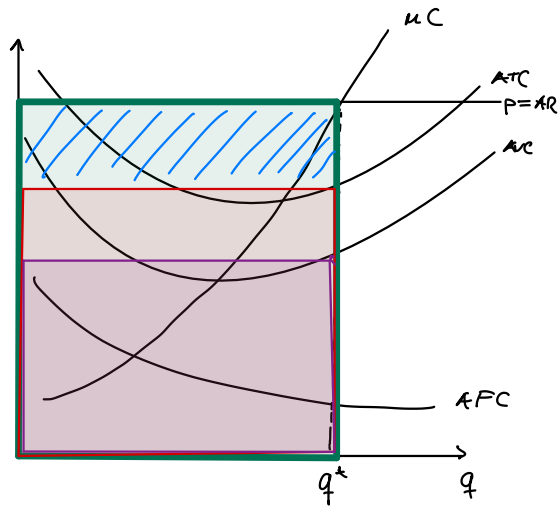
Total Cost : $q^+ ATC(q^+)$



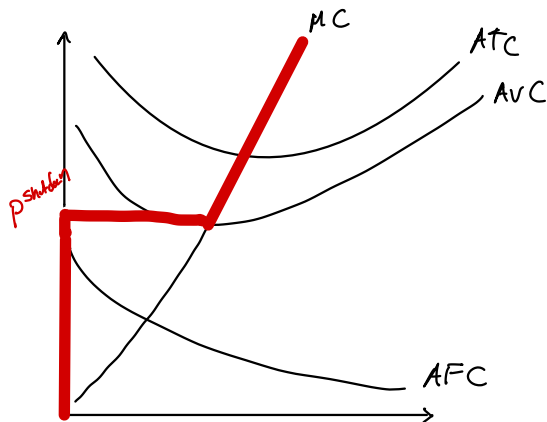
Profits



Variable cost : $q^+ AVC(q^+)$



Competition in the Short Run & the Short Run Output Decision



Short-Run Supply Curve of the firm:

$$p = MC(q), \text{ above}$$

p^{shutdown}

→ Inverse supply curve

Example: $MC(q) = c + dq$

$p^{\text{shutdown}} = c$. supply curve: $p = c + dq$

$$\Rightarrow q^{\text{supply}} = \frac{p - c}{d} \quad p \geq c = p^{\text{shutdown}}$$

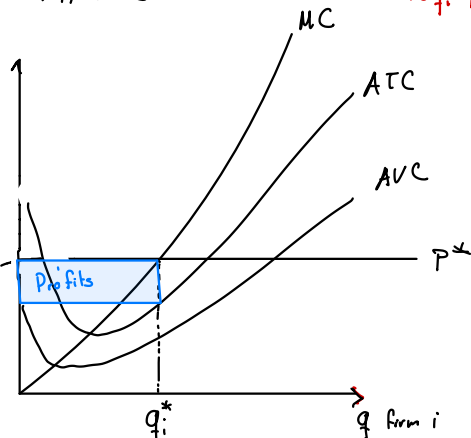
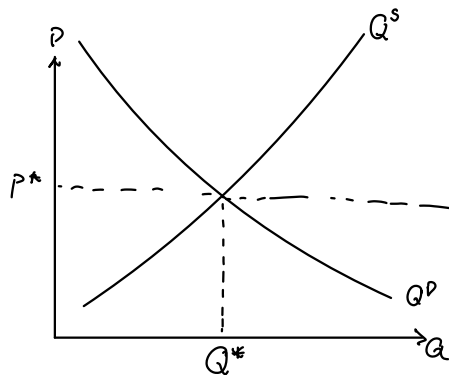
Competition in the Short Run & the Short Run Output Decision

For each firm i , we get $q_i^s(p) = q^s(p)$, all firm identical, $q_i^* = \frac{Q^*}{n}$

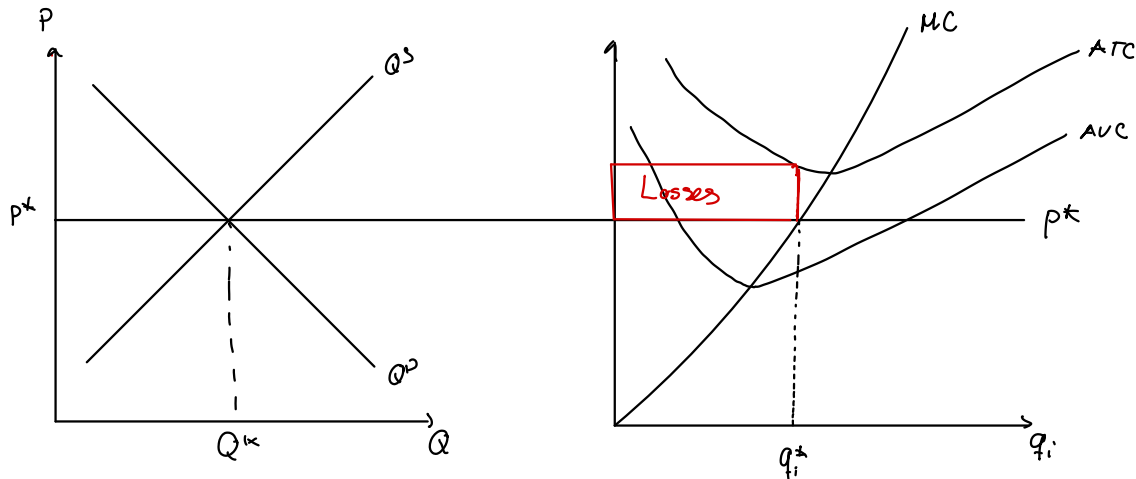
$Q^s(p) := \sum_i q_i^s(p)$ mkt. supply $\Rightarrow n q^s(p)$
 $\hookrightarrow \# \text{ firms}$

$Q^D(p) :=$ mkt. demand

$$n q_i^s(p^*) = Q^D(p^*)$$



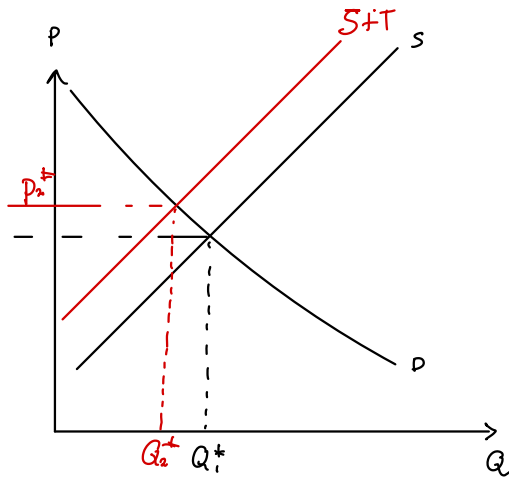
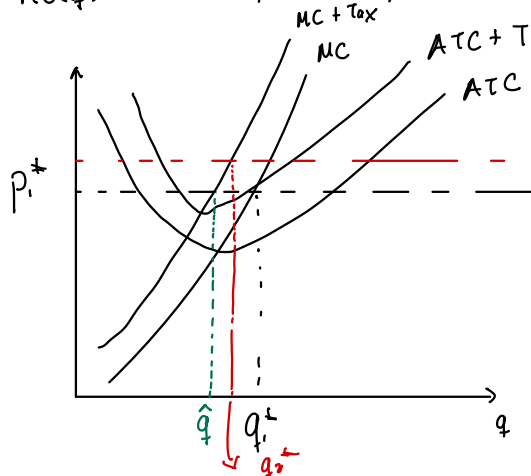
Competition in the Short Run & the Short Run Output Decision



Competition in the Short Run & the Short Run Output Decision

Suppose the gov. imposes a per-unit tax of T per unit

$$MC(q) \rightarrow MC^T(q) = MC(q) + T$$



Competition in the Short Run & and the Short Run Output Decision

$$Q^D(P) = \sum_{k=1}^K q_k^d(P)$$

$$Q^S(P) = \sum_{n=1}^N q_n^s(P)$$

(Q^*, P^*) such that

$$Q^D(P^*) = Q^* = Q^S(P^*)$$



look at the individual firm:

$$P^* = MC_i(q_i^s)$$

$$P^* \geq P_i^{\text{shutdown}}$$

In the Short-Run

Firms can be making:

- Profits : $P^* \geq \min ATC$
- Losses (if $P^* \geq \text{shutdown price of the firm}$)

Free entry
and
exit

Long-Run

Long-Run Market Equilibrium

Long-Run Equilibrium Condition: Zero (Economic) Profits

↳ SR : Profits $> 0 \rightarrow$ Entry
Profits $< 0 \rightarrow$ Exit

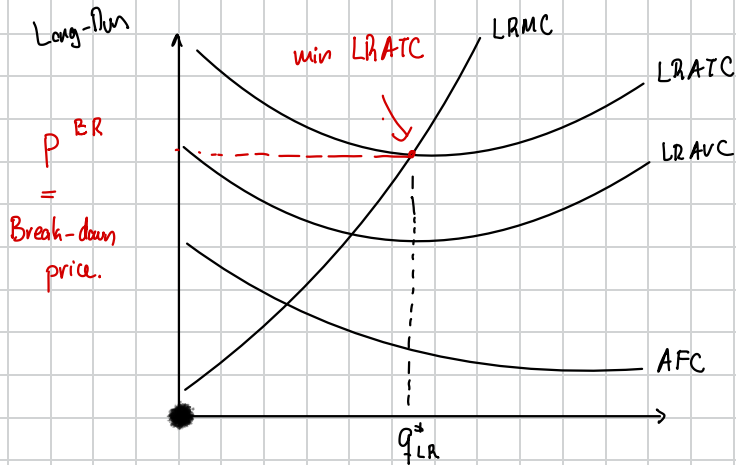
Long-Run : No cost is sunk (all costs avoidable)

↳ Can still have fixed costs, but are avoidable.

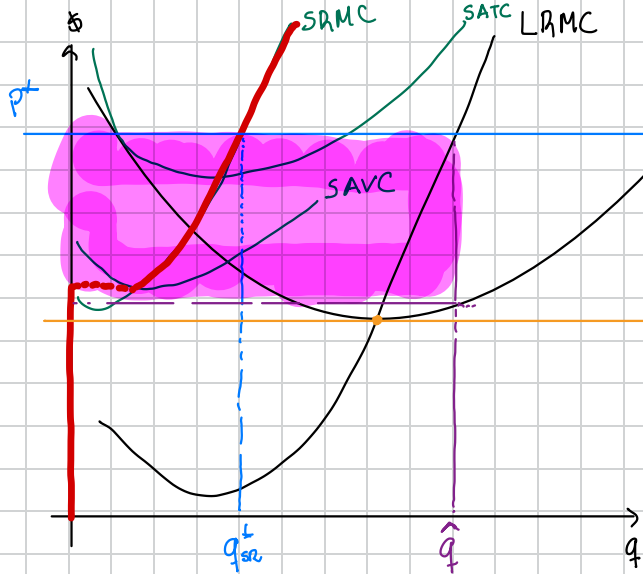
Long-Run cost function $c^u(0) = 0$

$c^{sr}(0) = F$ (you pay the fixed cost, if any)

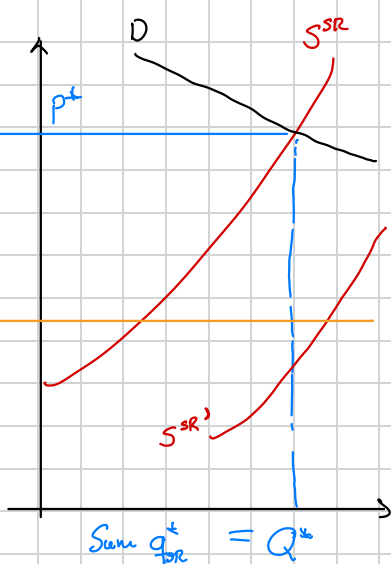
Long-Run Costs: Come from the full cost minimization problem of the firm.



P^{BR} in the Long-Run
determines the long
Run Equilibrium



Break-down
Price
= Long-Run
Equilibrium
Price.



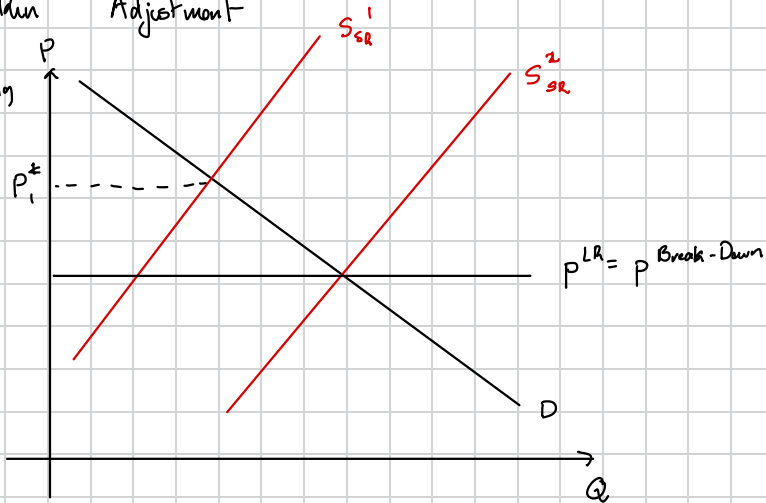
$\Rightarrow$ Firm is making profits is
 $C^{SR} = C^{LR}$

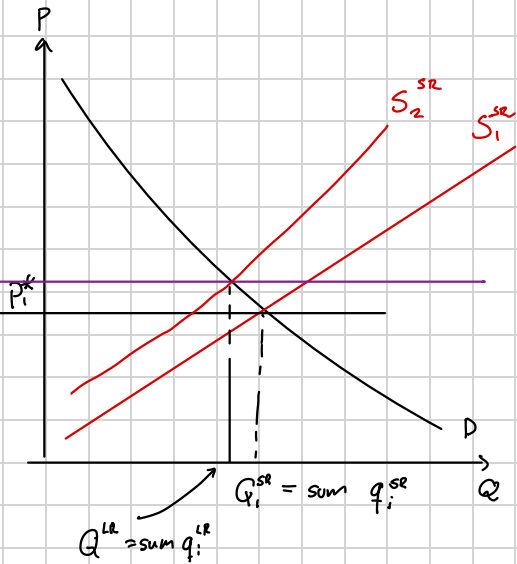
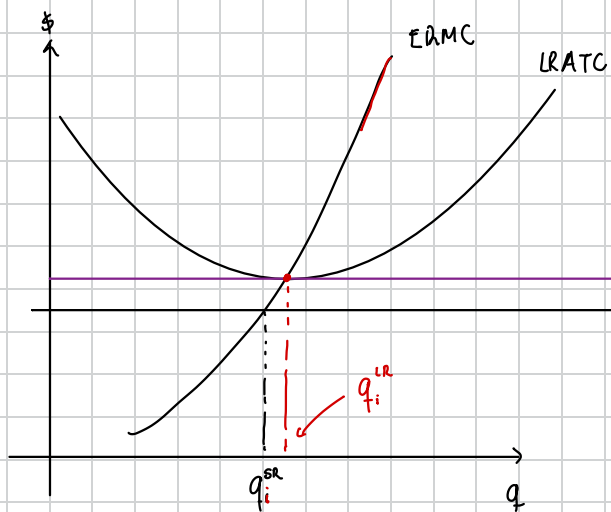
Long - Run

Adjustment

Firms
 $\pi > 0$
 $P_i \neq$

making
at
 $\rightarrow P_i \neq$



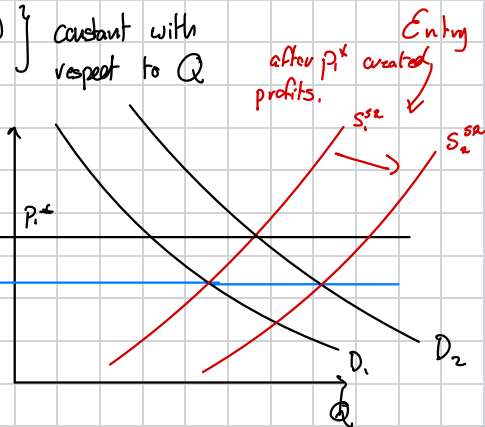
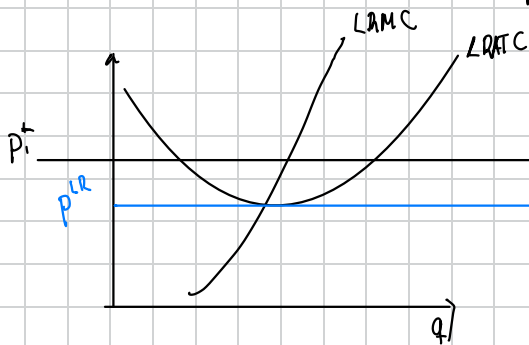


Long-Run Supply Curve

- Suppose all firms identical: Same Cost Structure

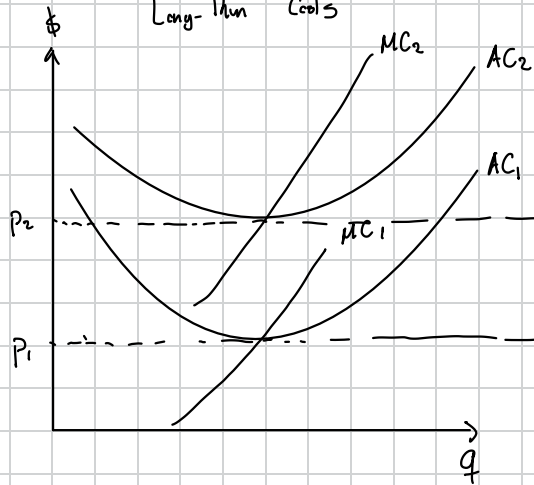
Industry with Constant Costs:

$\omega(Q)$
 $r(Q)$ } constant with respect to Q

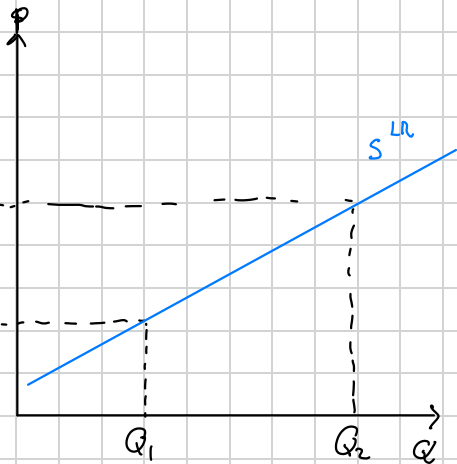


Industries with Increasing Costs

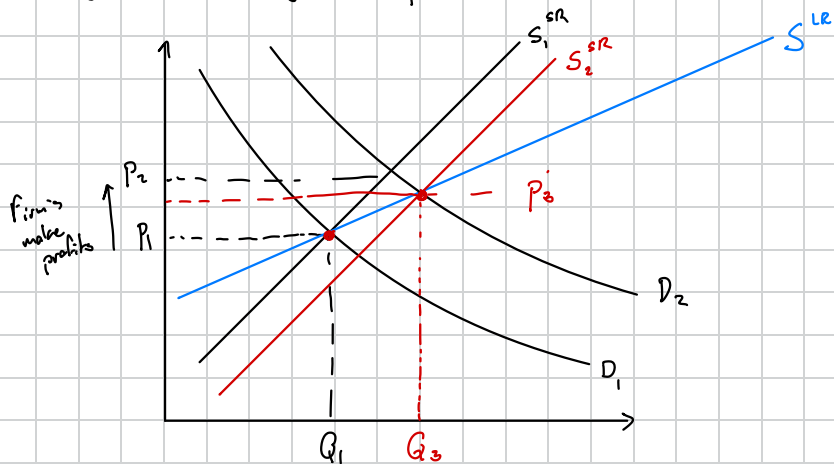
Long-run Costs



$w(Q) \uparrow$ increase with Q
 $r(Q) \uparrow$ increase with Q



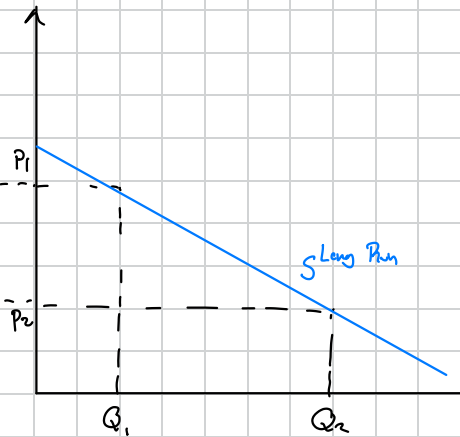
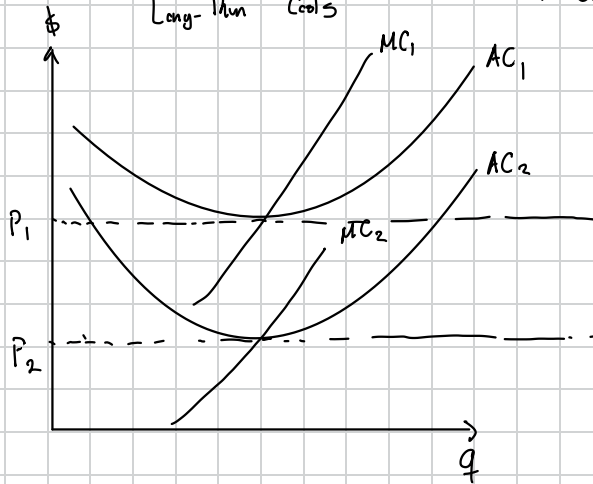
Increasing Costs Long Run eq.

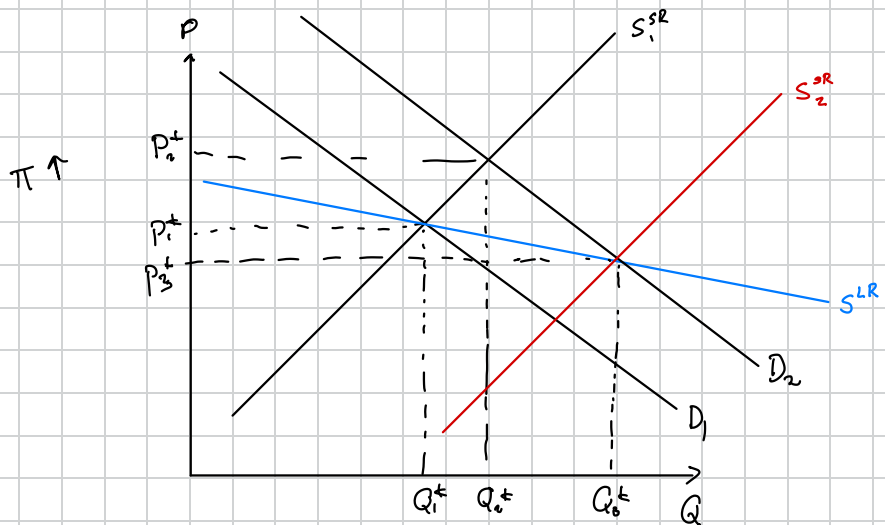


Decreasing Cost industry

Long-Run Costs

$w(Q) \downarrow$
 $r(Q) \downarrow$





Long-Run Competitive eq. example.

$$Q^D = 100,000 - 2P$$

Initially, there are 10,000 identical firms with Long-Run Cost

$$LTC(q) = 10,000 + 100q + 10q^2$$

$$LMC(q) = 100 + 20q$$

$$LAVC(q) = 100 + 10q$$

Shut-down $(q^s, p^s) \Rightarrow LMC = LAVC$

$$100 + 20q^s = 100 + 10q^s \Rightarrow q^s = 0$$

$$q^s = 0 \Rightarrow p^s = 100$$

Break-even $\Rightarrow (q^{BE}, p^{BE}) \Rightarrow LMC = LATC$
price

$$LATC(q) = \frac{10,000}{q} + 100 + 10q$$

$$\frac{10,000}{q^{BE}} + 100 + 10q^{BE} = 100 + 20q^{BE}$$

$$\sqrt{\frac{10,000}{10}} = q^{BE} \Rightarrow q^{BE} = \sqrt{1000} \approx 31.62$$

$$p^{BE} = \$732.45$$

"Initial" equilibrium is: Long-run supply curve:

$$P = MC \Rightarrow P = 100 + 20q \Rightarrow \frac{P-100}{20} = q^{SR}$$

$$Q_{SR}^s = 10,000 \left(\frac{P-100}{20} \right) = 500(P-100)$$

$$P^*: 500(P^*-100) = 100,000 - 2P^*$$

$$\Rightarrow 500P^* - 50,000 = 100,000 - 2P^*$$

$$\Rightarrow 502P^* = 150,000 \Rightarrow P_{SR}^* = \frac{150,000}{502} \approx 298.8$$

$Q_{SR}^* = 99,402.39$, each firm produces

$$q_{SR}^* = 9.940239$$

Are individual firms making profits?

$$\pi(q^*) = (P^* - LATC(q^*))q^* < 0 \quad \text{firms are making losses}$$

$$LATC(q^*) = 1205.41 > 298.8 = P^*$$

Long-Run example continued

Constant Cost Industry

Long-Run Competitive eq. example.

$$Q^D = 100,000 - 2P$$

Initially, there are 10,000 identical firms

with Long-Run Cost

$$LTC(q) = 10,000 + 100q + 10q^2$$

$$LMC(q) = 100 + 20q$$

$$LAVC(q) = 100 + 10q$$

$$\text{Shutdown } (q^S, p^S) \Rightarrow LMC = LAVC$$

$$100 + 20q^S = 100 + 10q^S \Rightarrow q^S = 0$$

$$q^S = 0 \Rightarrow p^S = 100$$

$$\text{Break-even} \Rightarrow (q^{BE}, p^{BE}) \Rightarrow LMC = LATC$$

$$LATC(q) = \frac{10,000}{q} + 100 + 10q$$

$$\frac{10,000}{q^{BE}} + 100 + 10q^{BE} = 100 + 20q^{BE}$$

$$\sqrt{\frac{10,000}{10}} = q^{BE} \Rightarrow q^{BE} = \sqrt{1,000} \approx 31.62$$

"Initial" equilibrium is: Long-run supply curve:

$$P = MC \Rightarrow P = 100 + 20q \Rightarrow \frac{P - 100}{20} = q$$

$$Q^S = 10,000 \left(\frac{P - 100}{20} \right) = 500(P - 100)$$

$$P^*: 500(P^* - 100) = 100,000 - 2P^*$$

$$\Rightarrow 500P^* - 50,000 = 100,000 - 2P^*$$

$$\Rightarrow 502P^* = 150,000 \Rightarrow P^* = \frac{150,000}{502} \approx 298.8$$

$$Q^* = 99,402.39, \text{ each firm produces}$$

$$q^* = 9,940.239$$

Are individual firms making profits?

$$\pi(q^*) = (P^* - LATC(q^*))q^* < 0 \quad \text{firms are making losses}$$

$$LATC(q^*) = 1205.41 > 298.8 = P^*$$

Firms are making losses, some firms will need to exit.

The Long-run eq. price is p^{BE}

$$\Rightarrow p^{BE} = LMC(q^{BE}) = 100 + 20\sqrt{1,000} \approx 732.4555$$

$$\Rightarrow p^{LR} = 732.4555 = S^{LR}$$

With PLR we compute

final Q^{LR} long-run eq. quantity:

- We use the demand because we still don't know how many firms need to exit:

$$Q^{LR} = 100,000 - 2P^{LR} = 98,535.088$$

$$N^{LR} = \frac{Q^{LR}}{q^{BE}} = \frac{98,535.088}{31.62} \approx 3115.9$$

$$\Rightarrow N^{LR} \approx 3116 \text{ firms}$$

need to be in the market, which means that from the original 10,000 firms, 6,884 leave the market.

